

# Simultaneous visible spectrophotometry of interstellar object 3I/ATLAS with Seimei/TriCCS

Jin BENIYAMA<sup>1,2,\*</sup>

<sup>1</sup>Université Côte d'Azur, Observatoire de la Côte d'Azur, CNRS, Laboratoire Lagrange, Bd de l'Observatoire, CS 34229, 06304 Nice Cedex 4, France

<sup>2</sup>The Department of Earth and Planetary Science, The University of Tokyo, 7-3-1 Hongo, Bunkyo, Tokyo 113-0033, Japan

\*E-mail: jbeniyama@oca.eu

ORCID: 0000-0003-4863-5577

## Abstract

3I/ATLAS, also known as C/2025 N<sub>1</sub> (ATLAS), is the third interstellar object (ISO) discovered in July 2025. ISOs are particularly interesting because characterizing their physical properties helps us understand and test our knowledge of Solar System formation. Several quick response observations of 3I/ATLAS were performed during the first few days after the discovery, and various results, such as reflectance spectra, have been reported. We performed simultaneous visible spectrophotometry of 3I/ATLAS from data taken using the TriColor CMOS Camera and Spectrograph (TriCCS) on the Seimei 3.8 m telescope. The Seimei/TriCCS observations of 3I/ATLAS were obtained in the  $g$ ,  $r$ ,  $i$ , and  $z$  bands in the Pan-STARRS system on UTC July 15, 2025. Our lightcurves show no significant variations during the 2.3 h observation, which is in good agreement with previous studies. Visible color indices of 3I/ATLAS,  $g - r = 0.603 \pm 0.031$ ,  $r - i = 0.210 \pm 0.031$ ,  $i - z = 0.117 \pm 0.046$ , and  $r - z = 0.327 \pm 0.035$  suggest it has a red surface similar to, or slightly redder than, that of D-type asteroids. Continuous observations of 3I/ATLAS before and after its perihelion passage in October 2025 are desired to investigate its physical properties.

**Keywords:** comets: general, comets: individual (C/2025 N<sub>1</sub>, 3I/ATLAS), techniques: photometric

arXiv:2508.08829v1 [astro-ph.EP] 12 Aug 2025

## 1 Introduction

Small bodies formed outside of the Solar System but are passing through it are called interstellar objects (ISO) or interstellar interlopers (e.g., Jewitt, Seligman 2023, for review). ISOs are particularly of interest because their physical characteristics can reveal valuable information about the early history of the Solar System. A comparative study of ISOs and Small Solar System Bodies (SSSBs) might allow us to investigate the planetesimal formation in different systems. The first ISO 1I/‘Oumuamua was discovered by the Pan-STARRS survey (Chambers et al. 2016) on October 19, 2017 (Meech et al. 2017), while the second ISO 2I/Borisov was discovered by G. Borisov on August 30, 2019. The two ISOs were quickly and extensively characterized following their discoveries, as such observations would be nearly impossible once the ISOs leave the Solar System. The surface colors of ISOs at visible to near-infrared wavelengths give insight into their composition and potential origins by comparing them with those of SSSBs (e.g., Meech et al. 2017; Ye et al. 2017; Jewitt et al. 2017; Bannister et al. 2017; Bolin et al. 2018; Fitzsimmons et al. 2018; Jewitt, Luu 2019; Guzik et al. 2020; Bolin et al. 2020). Since 2019, no ISOs had been discovered for about six years, and our knowledge of ISOs remained limited due to the small sample size.

On July 1, 2025, the third ISO 3I/ATLAS, also known as C/2025 N<sub>1</sub> (ATLAS), was discovered by the Asteroid Terrestrial-impact Last Alert System (ATLAS, Tonry et al. 2018) in Chile (Denneau et al. 2025). Soon after the discovery, weak cometary activity was reported by several observers across the world. Initial characterizations were performed worldwide using instruments such as the Canada-France-Hawaii Telescope (Seligman et al. 2025), the Very Large Telescope (Seligman et al. 2025; Opitom et al. 2025; Alvarez-Candal et al. 2025), Kottamia Astronomical Observatory 1.88-m telescope, Palomar 200-inch telescope, Apache Point Observatory Astrophysical Research Consortium 3.5-m telescope (Bolin et al. 2025; Belyakov et al. 2025), the 10.4 m Gran Telescopio Canarias (de la Fuente Marcos et al. 2025), the NASA Infrared Telescope Facility (IRTF, Karetta et al. 2025; Yang et al. 2025), the Gemini South telescope (Yang et al. 2025), and several smaller telescopes. Also, the 37 detections from the Vera C. Rubin Observatory were reported (Chandler et al. 2025). The results of existing observations are diverse, with different spectra reported, making interpretation challenging. Regarding its lightcurves, de la Fuente Marcos et al. (2025) reported a periodicity of  $16.79 \pm 0.23$  hours based on lightcurves obtained over approximately three days, while Seligman et al. (2025) concluded that no periodicity is apparent from observations over a period of approximately four days. Independent and reliable measurements are required before the perihelion passage in October 2025, when the comet could become more active.

In this letter, we present the results of simultaneous visible spectrophotometry of 3I/ATLAS. The observations were conducted on July 15, 2025, when 3I/ATLAS was at a heliocentric distance of 4.03 au. We describe the observations and data reduction in section 2, and present the results and discussion in section 3. We present the conclusion in section 4.

## 2 Observations

Visible spectrophotometric data of 3I/ATLAS were acquired on July 15, 2025. A summary of the observational conditions is provided in table 1. The predicted *V* band magnitudes in table 1 were

obtained from NASA JPL Horizons<sup>1</sup> using the Python package astroquery (Ginsburg et al. 2019). Observations of 3I/ATLAS were carried out with the TriColor CMOS Camera and Spectrograph (TriCCS) mounted on the 3.8 m Seimei Telescope (Kurita et al. 2020) under a single-epoch DDT program (PI: Jin Beniyama). The telescope is located at the Kyoto University Okayama Observatory (133.5967° E, 34.5769° N, and 355 m in altitude). Three-band imaging was simultaneously conducted using combinations of the Pan-STARRS (*g*, *r*, *i*) and (*g*, *r*, *z*) filters. The instrument covers a field of view of  $12.6' \times 7.5'$  with a pixel scale of 0.350 arcsec pixel<sup>-1</sup>.

At the time, 3I/ATLAS had a heliocentric distance of about 4.03 au and a geocentric distance of about 3.11 au. The phase angle of 3I/ATLAS was approximately 7.1 deg on July 15, 2025. The apparent sky motion of 3I/ATLAS was about 1.5 arcsec min<sup>-1</sup>. The percent of the Moon illuminated was about 76%, and the lunar elongation was about 87–88 deg. The sky was clear, and the seeing measured by using in-field stars was 2.2–2.6 arcsec in the *r* band.

We obtained images in the non-sidereal tracking mode during the observations of 3I/ATLAS. Also, we obtained sidereal tracking images to measure the seeing. Exposure times were set to 60 s for all observations. We performed standard image reduction, including bias subtraction, dark subtraction, and flat-fielding. The astrometry of in-field sources from the Gaia Data Release 2 was performed using the astrometry.net software (Lang et al. 2010).

To improve the signal-to-noise ratio (S/N) of 3I/ATLAS while avoiding the elongations of their images, we performed mean stacking prior to photometry, as shown in figure 1. Ten successive 60 s exposures were stacked to obtain images with an effective exposure times of 600 s. The two stacked frames, which overlap with the bright stars, were excluded from the analysis (see table 1). A typical readout time of the CMOS sensors on TriCCS is 0.4 milliseconds, which is negligibly small compared to the exposure time of 60 s. We also stacked frames using the World Coordinate System (WCS) of images corrected with the surrounding sources to suppress the elongations of the images of reference stars.

A custom pipeline to perform photometry using the same instrument was developed and used in the previous papers (Beniyama et al. 2023c, 2023b, 2023a, 2024, 2025). The field of 3I/ATLAS on that night was crowded with a galactic coordinate of  $(l, b) = (7.7^\circ, 7.7^\circ)$ . Thus, we developed and used a modified pipeline to analyze images of 3I/ATLAS. Circular aperture photometry was performed on both 3I/ATLAS and the reference stars using the SExtractor-based Python package sep. The local background estimated as a median value of the  $85 \times 85$  pix cutout image centered at the 3I/ATLAS or a reference star was subtracted before photometry. The aperture radii were set to 12 pix for reference stars, and were examined for 3I/ATLAS (see section 3). Photometry of 3I/ATLAS was conducted on non-sidereally stacked frames, while reference stars were measured using sidereal stacked frames.

All images were calibrated using Pan-STARRS catalog Data Release 2 (Chambers et al. 2016). Since the field was crowded, we manually selected reference stars that are far from any other sources. Extended sources, possible quasars as well as variable stars were removed using objinfoflag and objfilterflag in the Pan-STARRS catalog. In total, 15 reference stars that met the following criteria were used in the analysis:  $0.0 \leq (g-r)_{\text{PS}} \leq 1.1$  and  $0.0 \leq (r-i)_{\text{PS}} \leq 0.8$ , where  $(g-r)_{\text{PS}}$  and  $(r-i)_{\text{PS}}$  are colors

<sup>1</sup> <https://ssd.jpl.nasa.gov/horizons>

in the Pan-STARRS system. The typical uncertainty in the magnitude zero point is approximately 0.05 mag

### 3 Results and discussion

The lightcurves of 3I/ATLAS extracted from photometry using a fixed aperture of 6 pix, corresponding to approximately 5000 km from 3I/ATLAS, are shown in figure 2. We tested a range of aperture sizes from 5 to 12 pixels and found that the resulting relative brightnesses were nearly identical across this range, except for the largest apertures, where background contamination became non-negligible. Observations in the  $g$  and  $r$  bands spanned approximately 2.3 hours, yielding lightcurves that exhibit variations with amplitudes of roughly 0.3 mag. It might reflect the rotational variation due to changes in its cross-section. The amplitude of about 0.3 mag is slightly larger than the lightcurve amplitude of about 0.2 mag reported in de la Fuente Marcos et al. (2025), while a period of about 2.3 hours is notably short compared to the periodicity of  $16.79 \pm 0.23$  hours (de la Fuente Marcos et al. 2025). We identified possible contaminations from faint background sources near 3I/ATLAS by blinking successive images shown in figure 1, which corresponds to the lightcurve maxima in figure 2. Therefore, we interpret these brightness variations as likely artifacts rather than rotational variation. The relatively flat lightcurves of 3I/ATLAS are consistent with Seligman et al. (2025), which found no periodicity is apparent in the intensive time-series observations between July 2 and July 4, 2025. Kareta et al. (2025) also reported the absence of rotational variation from photometry spanning over three hours obtained with the IRTF. Additional lightcurve observations are crucial for constraining its rotational properties, particularly prior to its perihelion passage in October 2025, when 3I/ATLAS is expected to approach the Sun and potentially become more active.

As noted above, the lightcurve variations in figure 2 are likely instrumental artifacts, while they appear to be synchronized across all bands, suggesting that reliable color measurements can still be obtained. Colors of 3I/ATLAS were computed for each 600-second stacked frame. We examined a range of aperture sizes from 5 to 12 pixels and found the resulting color indices were identical within the uncertainties as shown in figure 3. We derived the median values and standard deviations of the color indices to be  $g-r = 0.603 \pm 0.031$ ,  $r-i = 0.210 \pm 0.031$ ,  $i-z = 0.117 \pm 0.046$ , and  $r-z = 0.327 \pm 0.035$  using a fixed aperture of 6 pix. In table 2, we summarize the derived colors of 3I/ATLAS with those reported in previous studies (Bolin et al. 2025; Seligman et al. 2025). We converted colors from Bolin et al. (2025) in the SDSS system into the Pan-STARRS system using the transformation equations from Tonry et al. (2012). We computed the propagated uncertainties of the colors in the Pan-STARRS system with the photometric errors and uncertainties in transformations.

In figure 4, we show the observed reflectance spectrum of 3I/ATLAS based on our visible spectrophotometry. The reflectances at the central wavelength of the  $r$ ,  $i$ , and  $z$  bands,  $R_r$ ,  $R_i$ , and  $R_z$ , were calculated as follows (e.g., DeMeo, Carry 2013):

$$R_r = 10^{-0.4[(r-g)_{3I} - (r-g)_{\odot}]}, \quad (1)$$

$$R_i = 10^{-0.4[(i-g)_{3I} - (i-g)_{\odot}]}, \quad (2)$$

$$R_z = 10^{-0.4[(z-g)_{3I} - (z-g)_{\odot}]}, \quad (3)$$

where  $(r-g)_{3I}$ ,  $(i-g)_{3I}$ , and  $(z-g)_{3I}$  are colors of 3I/ATLAS, whereas  $(r-g)_{\odot}$ ,  $(i-g)_{\odot}$ , and  $(z-g)_{\odot}$  are colors of the Sun in the Pan-STARRS system. We adopted solar magnitudes of  $g = 5.03$ ,  $r = 4.64$ ,  $i = 4.52$ , and  $z = 4.51$  from Willmer (2018),

and assumed an uncertainty of 0.02 in each. The reflectance spectra in figure 4 are normalized at the band center of  $g$  band in the Pan-STARRS system,  $0.4849 \mu\text{m}$  (Willmer 2018). Horizontal bars in 3I/ATLAS's spectrum indicate filter bandwidths. The calculated reflectance values are  $R_r$  of  $1.216 \pm 0.047$ ,  $R_i$  of  $1.321 \pm 0.063$ , and  $R_z$  of  $1.458 \pm 0.073$ . In figure 4, Mählke templates of S-, D-, and Z-type asteroids (Mählke et al. 2022) are shown for comparison. The reflectance spectra other than 3I/ATLAS are renormalized at the  $g$  band center. We found that the spectrum of 3I/ATLAS appears similar to that of D- or Z-type asteroids.

Our  $r-i$ ,  $i-z$  colors indices of 3I/ATLAS are consistent with those reported Seligman et al. (2025) and Bolin et al. (2025) within  $2\sigma$ . The  $r-z$  color index of 3I/ATLAS is also consistent with that in Seligman et al. (2025) within  $1.3\sigma$ , while slightly differs from that in Bolin et al. (2025) at  $2.2\sigma$  level. Our  $g-r$  color index of 3I/ATLAS differs from Seligman et al. (2025) and Bolin et al. (2025) at  $5.8\sigma$  and  $2.1\sigma$  levels, respectively. Our visible reflectance spectrum, which indicates a D-type asteroid surface, is largely in good agreement with the visible spectroscopy (Seligman et al. 2025; Opitom et al. 2025; de la Fuente Marcos et al. 2025; Yang et al. 2025).

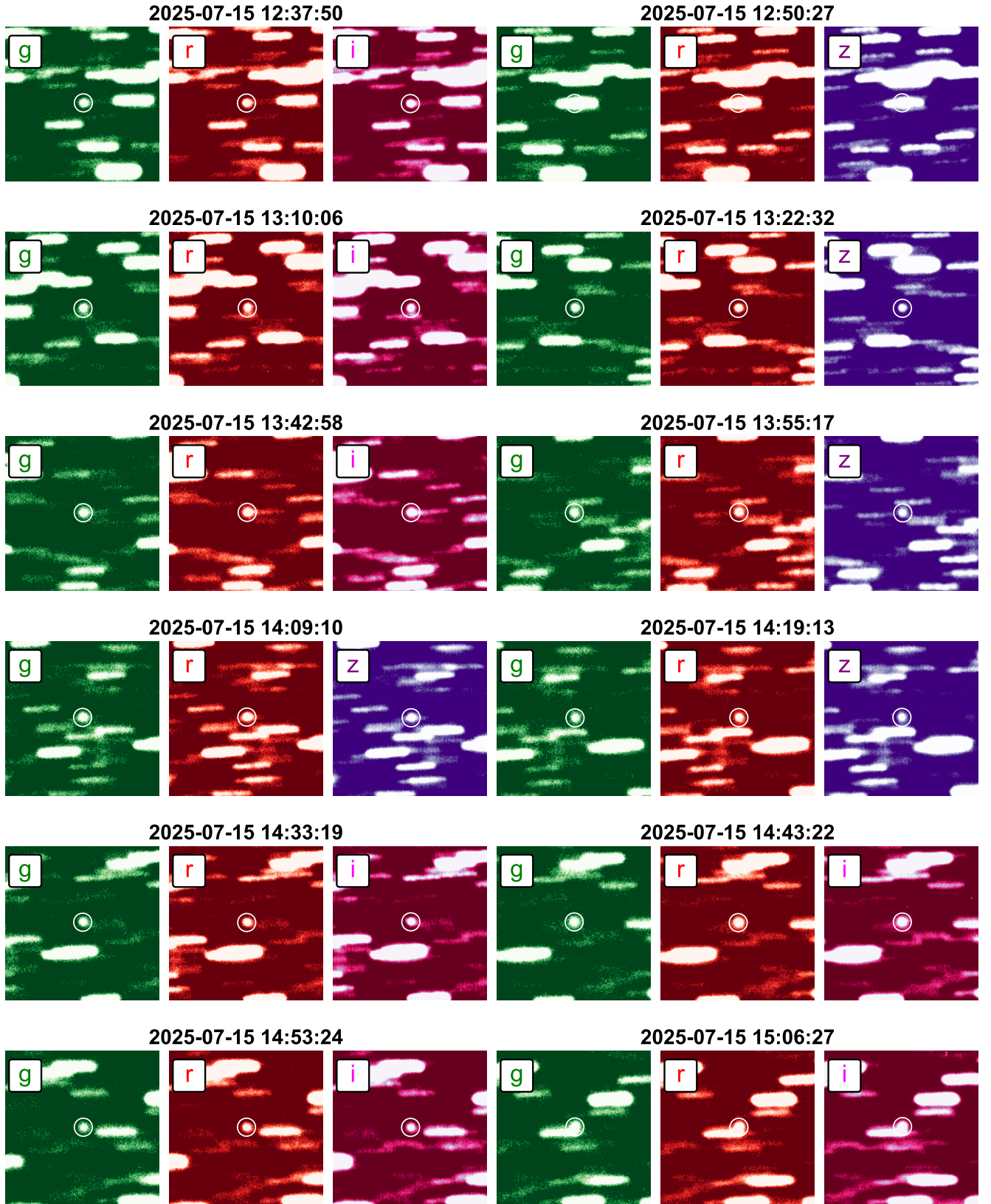
Discrepancies in visible colors have previously been reported for 1I/'Oumuamua (Meech et al. 2017; Jewitt et al. 2017; Bannister et al. 2017). Bannister et al. (2017) interpreted that the discrepancy indicates a potential surface color change, and this could also be the case for 3I/ATLAS. Apart from surface color changes, comet spectra may vary due to other mechanisms as well (Jewitt 2015, and references therein). Possible physical explanations include outbursts massive enough to affect the ambient environment (e.g., Yang et al. 2009) and chemically heterogeneous dust in the coma (Ivanova et al. 2017). 3I/ATLAS was at a relatively large heliocentric distance during both our observations and those reported previously (Seligman et al. 2025; Bolin et al. 2025), making significant outbursts unlikely. No strong activity has been reported to date. Further characterizations of 3I/ATLAS, especially prior to its perihelion passage in October 2025, will be essential to verify these results and investigate physical properties of the interstellar object.

### 4 Conclusion

We obtained simultaneous visible spectrophotometry of 3I/ATLAS using Seimei/TriCCS on July 15, 2025, when 3I/ATLAS was at a heliocentric distance of 4.03 au. The resulting lightcurves exhibit no significant brightness variation over a 2.3-hour time span, consistent with previous reports. We derived color indices for 3I/ATLAS of  $g-r = 0.603 \pm 0.031$ ,  $r-i = 0.210 \pm 0.031$ ,  $i-z = 0.117 \pm 0.046$ , and  $r-z = 0.327 \pm 0.035$  in the Pan-STARRS system, suggesting that the surface color of 3I/ATLAS closely resembles that of typical D-type asteroids. Further monitoring observations of 3I/ATLAS, especially before its perihelion passage in October 2025, will be essential to validate these results and to gain deeper insight into the physical properties of the interstellar object.

### Acknowledgments

We thank Dr. Akito Tajitsu, the Director of Subaru Telescope Okayama Branch Office, for the time allocation to observe 3I/ATLAS. We would like to thank Dr. Kenta Taguchi, and Dr. Masaru Kino, and Dr. Hideyuki Izumiura for observing assistance



**Fig. 1.** Non-sidereally stacked frames in *g*, *r*, *i*, and *z* bands with a total integration time of 600 s on July 15, 2025 are shown. Circles indicate 3I/ATLAS. Field of view covers 60'' × 60''. North is to the top, and east is to the left. Alt text: Non-sidereally stacked *g*, *r*, *i*, and *z* band images of 3I/ATLAS from July 15, 2025 (600 s total); 60'' × 60'' field of view; north is up, east is left; circles mark 3I/ATLAS.

**Table 1.** Summary of the observations

| Obs. Date<br>(UTC)   | Filter         | V<br>(mag) | Air Mass | Note                         |
|----------------------|----------------|------------|----------|------------------------------|
| 2025 Jul 15 12:37:50 | <i>g, r, i</i> | 17.4       | 1.68     |                              |
| 12:50:27             | <i>g, r, z</i> | 17.4       | 1.67     | Overlap with a bright source |
| 13:10:06             | <i>g, r, i</i> | 17.4       | 1.66     |                              |
| 13:22:32             | <i>g, r, z</i> | 17.4       | 1.67     |                              |
| 13:42:58             | <i>g, r, i</i> | 17.4       | 1.69     | Overlap with a faint source  |
| 13:55:17             | <i>g, r, z</i> | 17.4       | 1.71     |                              |
| 14:09:10             | <i>g, r, z</i> | 17.4       | 1.74     | Overlap with a faint source  |
| 14:19:13             | <i>g, r, z</i> | 17.4       | 1.77     |                              |
| 14:33:19             | <i>g, r, i</i> | 17.4       | 1.83     |                              |
| 14:43:22             | <i>g, r, i</i> | 17.3       | 1.87     | Overlap with a faint source  |
| 14:53:24             | <i>g, r, i</i> | 17.3       | 1.92     |                              |
| 15:06:27             | <i>g, r, i</i> | 17.3       | 2.01     | Overlap with a bright source |

\* Observation time in UT in midtime of exposure (Obs. Date), and filters (Filter) are listed.  
Predicted V band apparent magnitudes (V) at the observation starting time are referred to  
NASA Jet Propulsion Laboratory (JPL) Horizons as of July 22, 2025.  
Elevations of 3I/ATLAS to calculate air mass range (Air Mass) are  
also referred to NASA JPL Horizons.

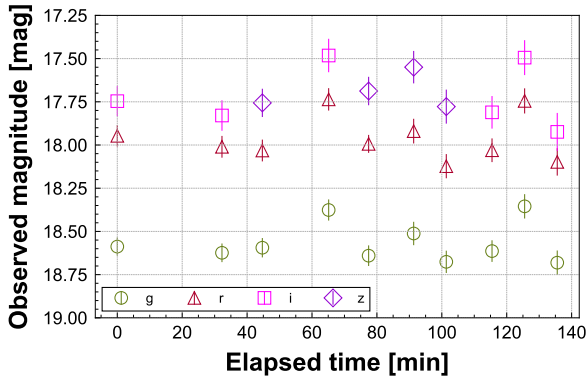
**Table 2.** Visible colors of 3I/ATLAS in the Pan-STARRS system

| Reference                        | Date           | Heliocentric dist.* | Aperture <sup>†</sup> | <i>g</i> − <i>r</i> | <i>r</i> − <i>i</i> | <i>i</i> − <i>z</i> | <i>r</i> − <i>z</i> |
|----------------------------------|----------------|---------------------|-----------------------|---------------------|---------------------|---------------------|---------------------|
| Seligman et al. (2025)           | 2025 July 2, 4 | 4.47, 4.40          | ~4000                 | 0.85 ± 0.03         | 0.25 ± 0.03         | 0.20 ± 0.08         | 0.45 ± 0.09         |
| Bolin et al. (2025) <sup>‡</sup> | 2025 July 6    | 4.33                | ~10000                | 0.717 ± 0.044       | 0.162 ± 0.030       | −0.018 ± 0.070      | 0.144 ± 0.076       |
| This study                       | 2025 July 15   | 4.03                | ~5000                 | 0.603 ± 0.031       | 0.210 ± 0.031       | 0.117 ± 0.046       | 0.327 ± 0.035       |

\* Heliocentric distance in au.

<sup>†</sup> Aperture radius from comet's nucleus in km.

<sup>‡</sup> Colors of 3I/ATLAS in the SDSS system reported in Bolin et al. (2025) were converted to those in the Pan-STARRS system using the equations in Tonry et al. (2012).



**Fig. 2.** Lightcurves of 3I/ATLAS. The observed *g*, *r*, *i*, and *z* bands magnitudes are presented as circles, triangles, squares, and diamonds, respectively. Bars indicate the  $1\sigma$  uncertainties. Alt text: Lightcurves of 3I/ATLAS in *g*, *r*, *i*, and *z* bands shown as circles, triangles, squares, and diamonds; error bars show  $1\sigma$  uncertainties.

using the Seimei telescope. We acknowledge the anonymous reviewer for the prompt and constructive comments that improved the quality of the paper. The authors thank the TriCCS developer team (which has been supported by the JSPS KAKENHI grant Nos. JP18H05223, JP20H00174, and JP20H04736, and by NAOJ Joint Development Research)

## Funding

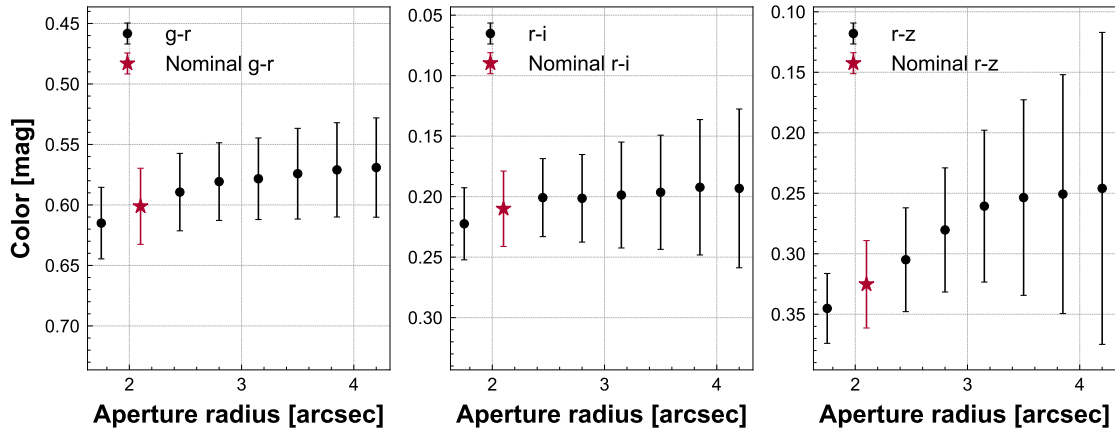
This work was supported by JSPS KAKENHI Grant Numbers JP23KJ0640 and 25H00665. This work was supported by the French government through the France 2030 investment plan managed by the National Research Agency (ANR), as part of the Initiative of Excellence Université Côte d'Azur under reference number ANR- 15-IDEX-01.

## Data availability

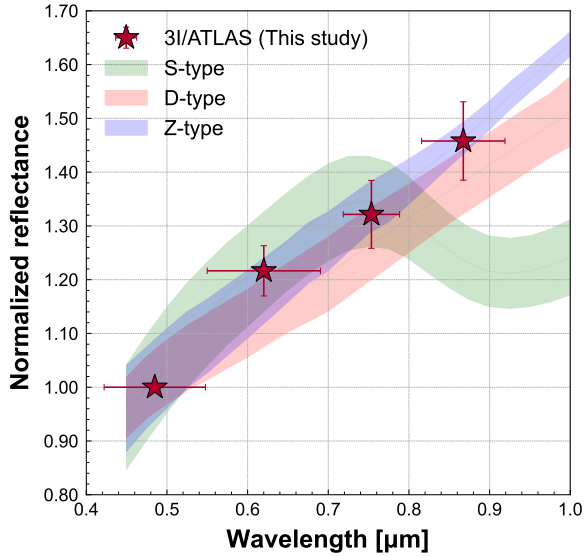
The data underlying this article are available at the following github repository [https://github.com/jinbeniyama/3I\\_Seimei](https://github.com/jinbeniyama/3I_Seimei).

## References

- Alvarez-Candal, A., Rizos, J. L., Lara, L. M., Santos-Sanz, P., Gutierrez, P. J., Ortiz, J. L., & Morales, N. 2025, arXiv e-prints
- Bannister, M. T., et al. 2017, *ApJL*, 851, L38
- Belyakov, M., et al. 2025, arXiv e-prints
- Beniyama, J., Bolin, B. T., Sergeyev, A. V., Delbo, M., Abron, L.-M., Belyakov, M., Sekiguchi, T., & Takagi, S. 2025, arXiv e-prints
- Beniyama, J., et al. 2023a, *AJ*, 166, 229
- Beniyama, J., et al. 2023b, *ApJ*, 955, 143
- Beniyama, J., et al. 2023c, *PASJ*, 75, 297
- Beniyama, J., Sergeyev, A. V., Tholen, D. J., & Micheli, M. 2024, *A&A*, 690, A180
- Bolin, B. T., et al. 2025, arXiv e-prints



**Fig. 3.** Color indices of 3I/ATLAS as a function of aperture radius. The median and standard deviation of the estimated colors are indicated by circles, while the nominal colors are marked with stars. Color indices remain nearly constant across aperture sizes from 5 to 12 pixels, indicating robustness of color measurement against aperture size. Alt text: Color indices ( $g-r$ ,  $r-i$ ,  $r-z$ ) of 3I/ATLAS vs. aperture radius. Circles indicate median values with standard deviation, stars indicate nominal values; colors remain stable from 5 to 12 pixels.



**Fig. 4.** Reflectance spectrum of 3I/ATLAS. Also, Mahlke templates of S-, D- and Z-type asteroids are shown (Mahlke et al. 2022). Shaded areas indicate the standard deviations of the template spectra. Alt text: Reflectance spectrum of 3I/ATLAS compared with S-, D-, and Z-type asteroid templates; shaded areas show  $1\sigma$  variations.

- Jewitt, D., & Luu, J. 2019, *ApJL*, 886, L29  
 Jewitt, D., Luu, J., Rajagopal, J., Kotulla, R., Ridgway, S., Liu, W., & Augustejn, T. 2017, *ApJL*, 850, L36  
 Jewitt, D., & Seligman, D. Z. 2023, *ARA&A*, 61, 197  
 Kareta, T., et al. 2025, arXiv e-prints  
 Kurita, M., et al. 2020, *PASJ*, 72, 48  
 Lang, D., Hogg, D. W., Mierle, K., Blanton, M., & Roweis, S. 2010, *AJ*, 139, 1782  
 Mahlke, M., Carry, B., & Mattei, P. A. 2022, *A&A*, 665, A26  
 Meech, K. J., et al. 2017, *Nature*, 552, 378  
 Opitom, C., et al. 2025, arXiv e-prints  
 Seligman, D. Z., et al. 2025, arXiv e-prints  
 Tonry, J. L., et al. 2018, *PASP*, 130, 064505  
 Tonry, J. L., et al. 2012, *ApJ*, 750, 99  
 Willmer, C. N. A. 2018, *ApJS*, 236, 47  
 Yang, B., Jewitt, D., & Bus, S. J. 2009, *AJ*, 137, 4538  
 Yang, B., Meech, K. J., Connelley, M., & Keane, J. V. 2025, arXiv e-prints  
 Ye, Q.-Z., Zhang, Q., Kelley, M. S. P., & Brown, P. G. 2017, *ApJL*, 851, L5

- Bolin, B. T., et al. 2020, *AJ*, 160, 26  
 Bolin, B. T., et al. 2018, *ApJL*, 852, L2  
 Chambers, K. C., et al. 2016, arXiv:1612.05560  
 Chandler, C. O., et al. 2025, arXiv e-prints  
 de la Fuente Marcos, R., et al. 2025, arXiv e-prints  
 DeMeo, F. E., & Carry, B. 2013, *ICARUS*, 226, 723  
 Denneau, L., et al. 2025, *Minor Planet Electronic Circulars*, 2025-N12  
 Fitzsimmons, A., et al. 2018, *Nature Astronomy*, 2, 133  
 Ginsburg, A., et al. 2019, *AJ*, 157, 98  
 Guzik, P., Drahos, M., Rusek, K., Waniak, W., Cannizzaro, G., & Pastor-Marazuela, I. 2020, *Nature Astronomy*, 4, 53  
 Ivanova, O., et al. 2017, *MNRAS*, 469, 2695  
 Jewitt, D. 2015, *AJ*, 150, 201